C-HI Full-grid Final Endpoint Quick-look

Date: 2026-06-07

Summary

This report records the first compact quick-look diagnostic for the completed full-grid, particles-enabled `C-HI` production endpoint on `kunanyi`.

Run facts:

- source production job: `697298.kunanyi-ohpc`
- production job name: `c_hi_r9_p64`
- scheduler result: `Exit_status = 0`
- resources: `select=64:ncpus=28:mpiprocs=28`
- MPI ranks: `1792`
- PBS walltime used: `07:49:58`
- restart mode: `RUN_KIND=h5_restart_9`
- PLUTO command: `-i pluto.ini -e environment.ini -maxtime 47.5 -h5restart 9`
- final PLUTO step: `49253`
- final PLUTO code time: `15330.06968940`
- physical time: `50.0000 Myr`, using `306.601394` PLUTO code units per Myr
- output tree volume reported by PBS stdout: `475G`

The apparently short production walltime is benign: PLUTO reached the configured `C-HI` endpoint at `tstop = 15330.06968940`, wrote the final HDF5 and particle products, and exited normally.

Endpoint Integrity

The production endpoint is:

`/scratch/bwbrook/xray_cavities_runs/2026-06-06-c-hi-fullgrid-agn-impi-r9-prod64/C-HI/particl`

Focused product-integrity checks under the AGN Intel/IMPI HDF5 stack found:

- final double output: `data.0011.dbl.h5`, `67318639276` bytes
- final float output: `data.0051.flt.h5`, `44890581676` bytes
- final particle file: `particles.5000.dbl`, `23608389` bytes
- both final HDF5 files contain `/vars/{prs,rho,tr1,vx1,vx2,vx3}` with shape `{1014, 960, 960}`
- scalar reads from `rho`, `vx1`, and `tr1` succeeded in both final HDF5 files
- the final particle header reports `196732` particles, time `1.533006968940e+04`, and step `49253`

The detailed integrity record is in `docs/exchange/logs/2026-06-07-c-hi-fullgrid-agn-impi-r9-prod64-i`

Quick-look Extraction

A one-core compute-node extraction job inspected the final float output:

- quick-look extraction job: `697386.kunanyi-ohpc`
- inspected file: `/scratch/bwbrook/xray_cavities_runs/2026-06-06-c-hi-fullgrid-agn-impi-r9-pr`
- timestep: `Timestep_51`
- extraction result: `Exit_status = 0`
- walltime: `00:04:34`
- peak memory: `3771068kb`
- requested envelope: `select=1:ncpus=1:mpiprocs=1:mem=32gb`, `walltime=01:00:00`

The compact evidence bundle was retrieved to:

`build/science/kunanyi_stage_2026-06-07/c_hi_fullgrid_final_quicklook_697386_compact_2026-06-`

SHA-256:

`26f8ab57e4f8e3913bc629abb391f699c75b45296d474fe2c411736a0b71098c`

Small extracted metadata and PBS logs are stored with the figures:

- `docs/science/assets/c_hi_fullgrid_final_quicklook_2026-06-07/metadata.txt`
- `docs/science/assets/c_hi_fullgrid_final_quicklook_2026-06-07/summary_metrics.csv`
- `docs/science/assets/c_hi_fullgrid_final_quicklook_2026-06-07/quicklook_pbs_stdout.txt`
- `docs/science/assets/c_hi_fullgrid_final_quicklook_2026-06-07/quicklook_pbs_stderr.txt`

Tracer Extent

At the final endpoint, tracer-bearing material spans most of the jet-axis domain but remains inside the apparent full-grid boundary in the quick-look slice. The default Level 2 truth threshold remains `tracer > 5e-3`.

Threshold	Cells	x range (kpc)	y range (kpc)	z range (kpc)
`1e-3`	194674862	-30.832 to 31.400	-32.861 to 31.115	-217.440 to 217.440
`5e-3`	185000464	-29.998 to 30.832	-32.269 to 30.551	-217.440 to 217.440
`1e-2`	177856594	-29.724 to 29.724	-31.977 to 30.273	-215.607 to 215.607

For comparison, the previous 5 Myr quick-look had a default-threshold extent of about `x,y = +/-9.46 kpc` and `z = +/-33.0 kpc`.

Figures

Full-domain mid-plane maps:

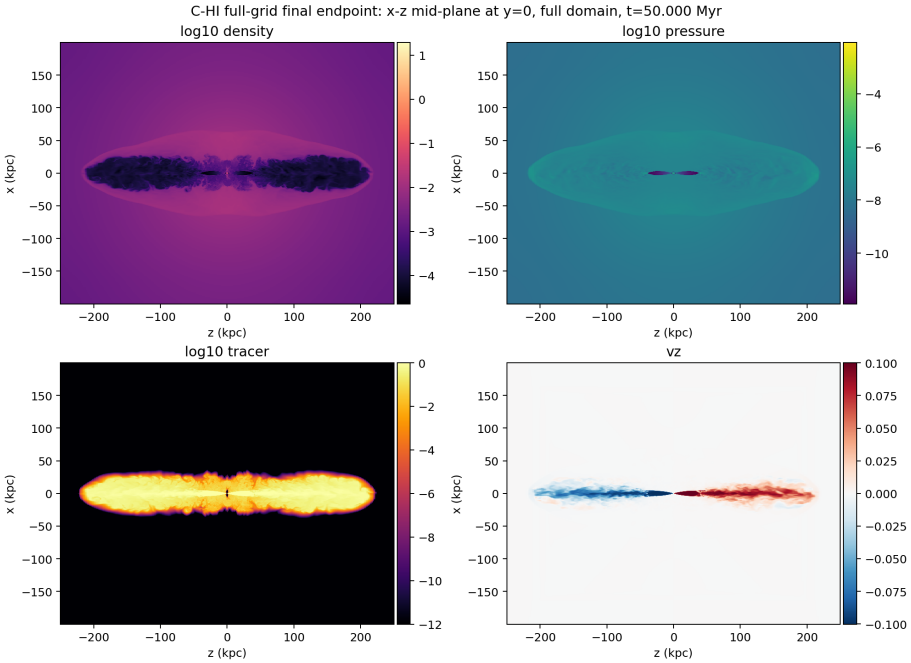


Figure 1: C-HI final full-domain quick-look maps

Central zoom:

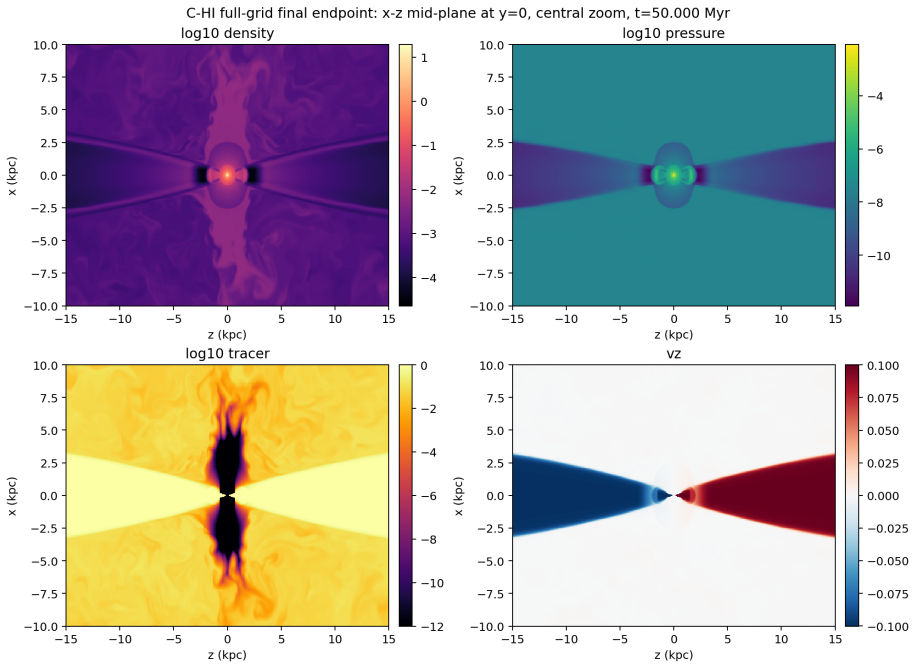


Figure 2: C-HI final central quick-look maps

Jet-axis profiles:

Interpretation

This endpoint passes the immediate production-readiness gate:

- the 64-node AGN Intel/IMPI restart stack reached the configured 50 Myr stop cleanly
- final restart-capable, float, XMF, particle, and catalogue products are present and readable
- the compact extractor can read the final full-grid HDF5 product on a compute node without special handling beyond a modest memory request
- the mid-plane maps are nonblank and show the expected bilateral jet/cocoon structure
- the default-threshold tracer extent is large but not obviously boundary clipped in this quick-look

This remains a run-health and readiness diagnostic, not a cavity `pV`, X-ray, radio, or lobe-energy measurement. The next science-facing step is to promote this endpoint into the aligned truth, X-ray, radio, and cavity-analysis workflow.

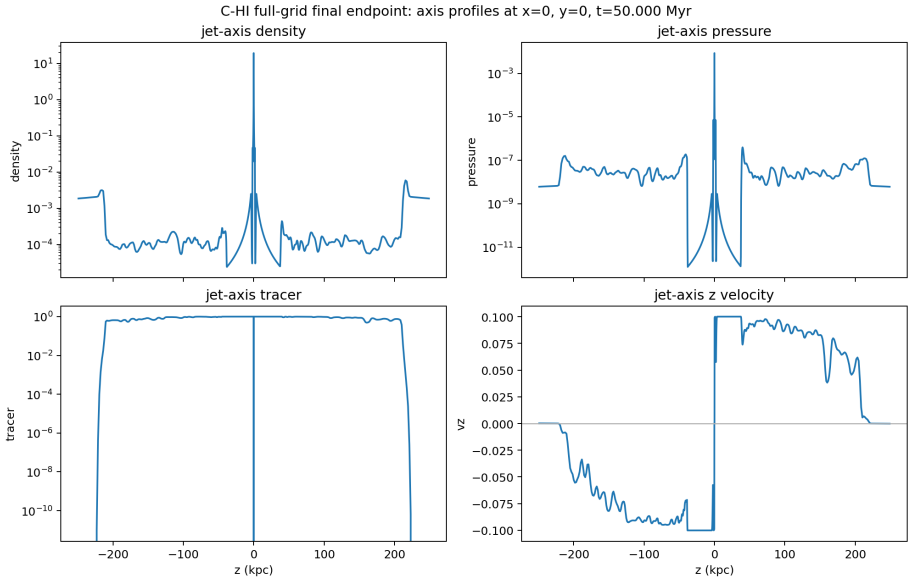


Figure 3: C-HI final quick-look axis profiles